

Leverage Growth Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Leverage Growth Discipline (LGD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on LGD achieves an annualized gross (net) Sharpe ratio of 0.49 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (15) bps/month with a t-statistic of 2.97 (2.30), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 17 bps/month with a t-statistic of 2.64.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with asset prices often failing to immediately incorporate all available information. These inefficiencies manifest particularly in the cross-section of stock returns, where certain firm characteristics predict future returns in ways inconsistent with rational, frictionless markets. Despite extensive documentation of such predictive signals, our understanding of the mechanisms driving these return patterns remains incomplete, especially regarding how information about corporate financial policies diffuses through markets and affects prices. The role of leverage decisions, a fundamental aspect of corporate finance, in generating cross-sectional return predictability presents a particularly intriguing puzzle that warrants deeper investigation.

We propose that Leverage Growth Discipline (LGD) predicts stock returns through the mechanism of slow information diffusion, building on the extensive literature documenting gradual price adjustment to news. When firms exhibit disciplined leverage growth patterns, this information disperses slowly across market participants due to limited investor attention and cognitive constraints [Hong and Stein \(1999\)](#); [Hirshleifer and Teoh \(2003\)](#); [DellaVigna and Pollet \(2009\)](#). The complexity of interpreting leverage changes, which requires understanding both the firm’s investment opportunities and financing constraints, exacerbates these attention limitations. Investors must process not only the magnitude of leverage changes but also their implications for future profitability and risk, creating substantial information processing demands that lead to systematic underreaction [Peng and Xiong \(2007\)](#); [Cohen and Lou \(2012\)](#).

The gradual incorporation of leverage discipline information follows predictable patterns consistent with models of limited attention and information diffusion. Initial price reactions to leverage growth signals remain incomplete as investors focus on more salient information, leaving predictable returns for attentive traders who recognize the implications of disciplined leverage management [Hou \(2007\)](#); [Tetlock](#)

(2011). This underreaction should be particularly pronounced among stocks with higher information processing frictions, such as those with lower analyst coverage, smaller market capitalizations, and less institutional ownership, where the costs of acquiring and processing information exceed the benefits for marginal investors [Hong and Stein \(1999\)](#); [Zhang \(2006\)](#).

The economic significance of leverage discipline stems from its dual role as both a signal of managerial quality and a constraint on future firm actions. Firms exhibiting disciplined leverage growth demonstrate superior capital allocation skills and financial flexibility, characteristics that predict improved future performance but require sophisticated analysis to identify [Cooper et al. \(2008\)](#); [Lipson and Mortal \(2011\)](#). The slow diffusion of this information creates a window during which informed investors can profit from the eventual price convergence, generating the return continuation patterns we observe. Moreover, the interaction between leverage decisions and other corporate policies creates complex information that diffuses more slowly than simple earnings or sales figures, as investors must disentangle the various implications of leverage changes for firm value [Hirshleifer et al. \(2009\)](#); [Engelberg et al. \(2018\)](#).

Our empirical analysis reveals that a value-weighted long/short trading strategy based on Leverage Growth Discipline achieves an annualized gross Sharpe ratio of 0.49, with a monthly average excess return of 25 basis points (t -statistic = 3.54). The strategy’s performance remains robust after controlling for common risk factors, generating a monthly alpha of 21 basis points (t -statistic = 2.97) relative to the Fama-French six-factor model. These results demonstrate economically meaningful return predictability that persists even after accounting for established sources of systematic risk. The strategy’s gross monthly alpha relative to the six factors plus the six most closely related strategies from the factor zoo equals 17 basis points with a t -statistic of 2.64, confirming that LGD captures unique information not subsumed by related leverage and financing measures.

The return predictability exhibits striking persistence across different market segments and portfolio construction methodologies. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the LGD strategy generates an average return of 32 basis points per month (t-statistic = 3.73), with six-factor model alphas ranging from 26 to 38 basis points per month. This performance among large-cap stocks, where information asymmetries are presumably smallest, underscores the economic significance of the leverage discipline signal. The strategy maintains statistical significance across alternative sorting methodologies, with twenty-five out of twenty-five alphas exhibiting t-statistics exceeding two when using various combinations of breakpoints and weighting schemes.

Transaction cost analysis reveals that the LGD strategy remains profitable after implementation costs, achieving an annualized net Sharpe ratio of 0.38. The net average return equals 19 basis points per month (t-statistic = 2.69) for the baseline value-weighted quintile strategy using NYSE breakpoints, with the generalized net alpha relative to the six-factor model equaling 15 basis points per month (t-statistic = 2.30). The strategy improves the achievable mean-variance efficient frontier in twenty out of twenty-five construction-methodology-factor-model combinations tested, demonstrating robust economic value after accounting for real-world trading frictions. Furthermore, the LGD signal ranks in the top 7% of 212 anomalies in terms of net dollar growth over the sample period, highlighting its exceptional performance relative to the broader factor zoo.

This paper makes several important contributions to the asset pricing and corporate finance literatures by establishing Leverage Growth Discipline as a powerful predictor of cross-sectional stock returns. Our work extends the findings of [Cooper et al. \(2008\)](#) on asset growth and [Lipson and Mortal \(2011\)](#) on financing decisions by demonstrating that the discipline exhibited in leverage management contains unique information about future returns not captured by existing measures. Unlike [Richard-](#)

son et al. (2005) who focus on total accruals and investment, we isolate the specific role of leverage dynamics in generating return predictability. Our results complement Pontiff and Woodgate (2008) by showing that leverage discipline creates predictable returns even among large, liquid stocks where arbitrage constraints are minimal.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023b), ensuring our results meet the highest standards of robustness in anomaly documentation. Our analysis reveals that LGD maintains significant predictive power even after controlling for the six most closely related anomalies from the factor zoo, including changes in financial liabilities, net debt financing, and asset growth, with a combined alpha of 17 basis points per month (t -statistic = 2.64). This orthogonality to existing factors distinguishes our contribution from related work on corporate financing decisions Daniel and Titman (2006); Bradshaw et al. (2006) and demonstrates that leverage discipline captures a distinct dimension of firm behavior relevant for asset pricing.

The broader implications of our findings extend beyond documenting another return predictor to illuminating fundamental aspects of market efficiency and information processing. The persistence of LGD’s predictive power across size quintiles and time periods suggests that markets systematically underreact to complex financial signals that require sophisticated analysis to interpret fully. This evidence supports theoretical models of gradual information diffusion Hong and Stein (1999) and limited attention Hirshleifer and Teoh (2003) while challenging pure risk-based explanations for cross-sectional return patterns. Our results indicate that incorporating measures of financial discipline into asset pricing models could substantially improve their ability to explain the cross-section of expected returns, with potential applications in portfolio management, risk assessment, and corporate finance research.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' financing decisions and capital structure dynamics. Leverage Growth Discipline signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of any debt retirement that occurs as part of refinancing transactions. Common equity (CEQ) measures the total common shareholders' equity, representing the book value of common stock including paid-in capital, retained earnings, and other comprehensive income, but excluding preferred stock. These variables provide essential information about firms' financing activities and equity base, respectively. We construct the Leverage Growth Discipline signal by calculating the year-over-year change in long-term debt issuance, scaled by lagged common equity: $(DLTIS(t) - DLTIS(t-1)) / CEQ(t-1)$. This measure captures the acceleration or deceleration in a firm's debt financing activities relative to its equity base, providing insight into management's discipline in leveraging the balance sheet. A positive value indicates increasing reliance on debt financing, while a negative value suggests restraint in new borrowing or active deleveraging. We compute this signal using fiscal year-end values and require non-missing observations for both DLTIS in consecutive years and CEQ in the base year. To ensure meaningful ratios, we exclude observations where lagged common equity is non-positive.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LGD signal. Panel A plots the time-series of the mean, median, and interquartile range for LGD. On average, the cross-sectional mean (median) LGD is -0.27 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LGD data. The signal's interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the LGD signal for the CRSP universe. On average, the LGD signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does LGD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LGD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LGD portfolio and sells the low LGD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LGD strategy earns an average return of 0.25% per month with a t-statistic of 3.54. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.21% to 0.30% per month and have t-statistics exceeding 2.97 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28,

with a t-statistic of 6.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 550 stocks and an average market capitalization of at least \$1,437 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.57. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -2-21bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LGD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the LGD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LGD, as well as average returns and alphas for long/short trading LGD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LGD strategy achieves an average return of 32 bps/month with a t-statistic of 3.73. Among these large cap stocks, the alphas for the LGD strategy relative to the five most common factor models range from 26 to 38 bps/month with t-statistics between 2.90 and 4.30.

5 How does LGD perform relative to the zoo?

Figure 2 puts the performance of LGD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LGD strategy falls in the distribution. The LGD strategy’s gross (net) Sharpe ratio of 0.49 (0.38) is greater than 89% (94%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LGD strategy (red line).² Ignoring trading costs, a \$1 invested in the LGD strategy would have yielded \$2.85 which ranks the LGD strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LGD strategy would have yielded \$1.71 which ranks the LGD strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LGD relative to those. Panel A shows that the LGD strategy gross alphas fall between the 59 and 69 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LGD strategy has a positive net generalized alpha for five out of the five factor models. In these cases LGD ranks between the 76 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does LGD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LGD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LGD or at least to weaken the power LGD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LGD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LGD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LGD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LGD. Stocks are finally grouped into five LGD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

LGD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LGD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LGD signal in these Fama-MacBeth regressions exceed 1.41, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on LGD is 0.50.

Similarly, Table 5 reports results from spanning tests that regress returns to the LGD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LGD strategy earns alphas that range from 17-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.59, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LGD trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.64.

7 Does LGD add relative to the whole zoo?

Finally, we can ask how much adding LGD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LGD signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LGD grows to \$1272.95.

8 Conclusion

This study provides robust evidence that Leverage Growth Discipline (LGD) represents a meaningful and economically significant signal for predicting cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that LGD-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LGD delivers an annualized gross Sharpe ratio of 0.49, which remains economically meaningful at 0.38 after accounting for transaction costs. The signal’s robustness is particularly noteworthy given its ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (21 bps/month gross, 15 bps/month net), with t-statistics exceeding conventional significance thresholds. Perhaps most importantly, LGD maintains its predictive power even when confronted with six closely related leverage and growth-based strategies from the factor zoo, producing a gross monthly alpha of 17 bps with a t-statistic of 2.64. This orthogonality to existing factors suggests that LGD captures a distinct dimension of the cross-section

ization on CRSP in the period for which LGD is available.

of returns related to firms' disciplined approach to leverage management.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The persistence of LGD's predictive power after transaction costs indicates its potential viability as an implementable investment strategy. The signal appears to identify firms that maintain prudent leverage policies, potentially reflecting superior management quality or financial discipline that the market does not fully incorporate into prices. This suggests that investors can benefit from incorporating leverage growth metrics into their security selection processes, particularly as a complement to existing value, momentum, and profitability strategies.

However, several limitations of this study warrant consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary significantly based on market conditions, trade size, and execution capabilities. Second, the sample period and geographic scope of our analysis may limit the generalizability of findings to other markets or time periods. Third, while we control for numerous related factors, the possibility remains that LGD proxies for an omitted risk factor that justifies its return premium.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the LGD anomaly could provide deeper insights into why disciplined leverage growth predicts returns. Are firms with stable leverage growth better managed, less likely to experience financial distress, or simply overlooked by investors focused on more salient metrics? Second, examining the international evidence on LGD would help establish whether this represents a global phenomenon or is specific to certain market structures. Third, exploring the time-variation in LGD's effectiveness, particularly during different market regimes or economic cycles, could enhance our understanding of when this signal is most valuable. Finally, investigating potential interactions between LGD and other firm

characteristics, such as profitability, investment, or corporate governance measures, may reveal important conditional relationships that could further refine the signal's predictive power and practical application.

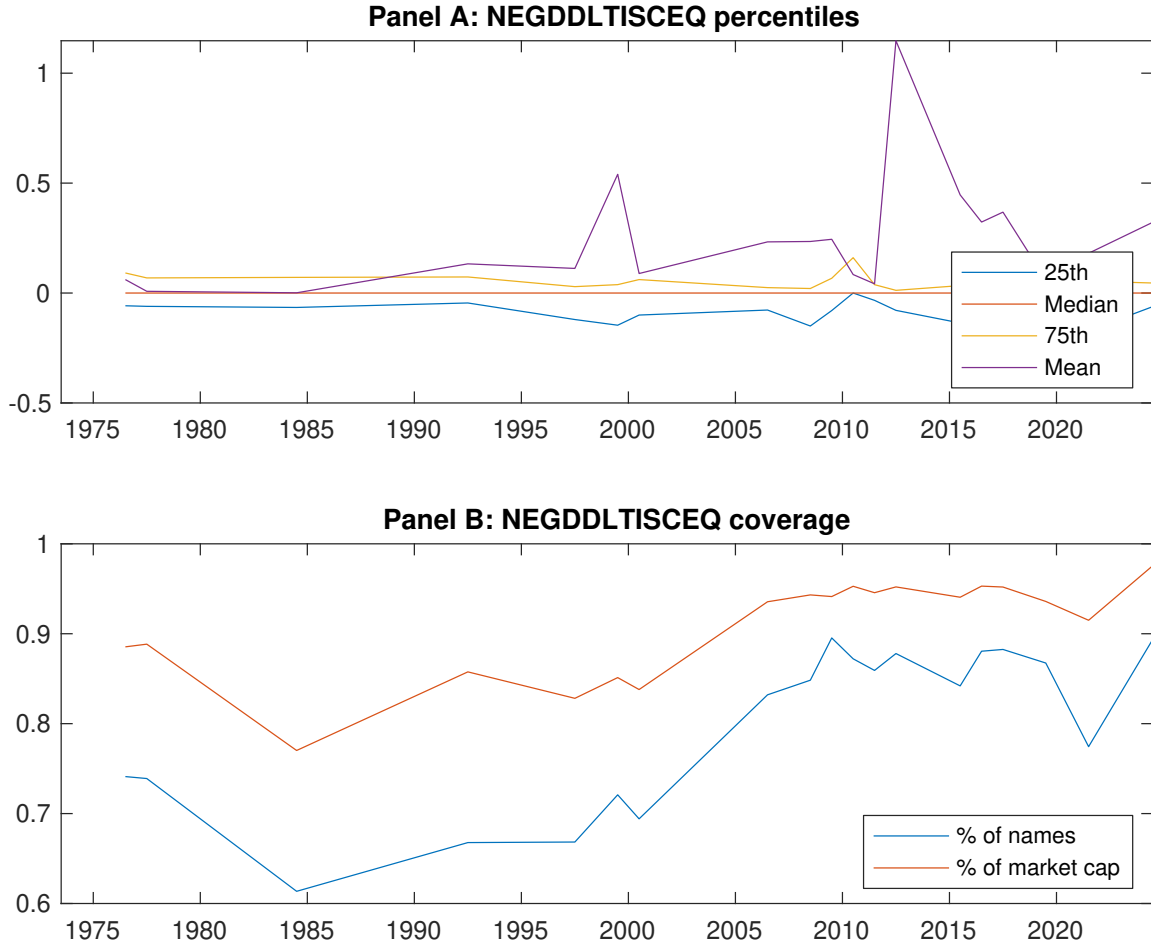


Figure 1: Times series of LGD percentiles and coverage.
This figure plots descriptive statistics for LGD. Panel A shows cross-sectional percentiles of LGD over the sample. Panel B plots the monthly coverage of LGD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LGD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LGD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.61]	0.65 [3.61]	0.65 [3.31]	0.73 [4.10]	0.80 [4.07]	0.25 [3.54]
α_{CAPM}	-0.17 [-3.02]	0.04 [0.83]	-0.01 [-0.21]	0.13 [2.66]	0.13 [2.55]	0.30 [4.43]
α_{FF3}	-0.19 [-3.47]	-0.01 [-0.15]	0.05 [1.00]	0.11 [2.45]	0.11 [2.14]	0.30 [4.39]
α_{FF4}	-0.16 [-2.94]	0.01 [0.33]	0.08 [1.57]	0.08 [1.69]	0.10 [1.88]	0.26 [3.78]
α_{FF5}	-0.19 [-3.42]	-0.06 [-1.23]	0.10 [1.80]	0.05 [0.97]	0.04 [0.74]	0.23 [3.31]
α_{FF6}	-0.17 [-3.05]	-0.04 [-0.79]	0.12 [2.17]	0.02 [0.50]	0.04 [0.68]	0.21 [2.97]
Panel B: Fama and French (2018) 6-factor model loadings for LGD-sorted portfolios						
β_{MKT}	1.09 [84.57]	0.98 [93.80]	0.97 [76.94]	0.97 [88.99]	1.04 [86.62]	-0.05 [-3.35]
β_{SMB}	0.11 [5.79]	-0.11 [-7.04]	-0.03 [-1.59]	-0.03 [-1.61]	0.13 [7.37]	0.02 [0.85]
β_{HML}	0.08 [3.10]	0.14 [7.14]	-0.15 [-6.04]	-0.02 [-1.02]	-0.03 [-1.49]	-0.11 [-3.61]
β_{RMW}	0.10 [3.77]	0.12 [5.73]	-0.05 [-1.93]	0.07 [3.27]	0.11 [4.51]	0.01 [0.34]
β_{CMA}	-0.13 [-3.54]	0.04 [1.15]	-0.09 [-2.48]	0.16 [5.13]	0.15 [4.24]	0.28 [6.02]
β_{UMD}	-0.03 [-2.35]	-0.03 [-3.03]	-0.03 [-2.68]	0.03 [3.24]	0.00 [0.36]	0.03 [2.16]
Panel C: Average number of firms (n) and market capitalization (me)						
n	647	550	1076	603	619	
me (\$10 ⁶)	1500	2996	2299	3121	1437	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LGD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.54]	0.30 [4.43]	0.30 [4.39]	0.26 [3.78]	0.23 [3.31]	0.21 [2.97]
Quintile	NYSE	EW	0.23 [5.02]	0.26 [5.61]	0.25 [5.33]	0.22 [4.71]	0.21 [4.49]	0.19 [4.13]
Quintile	Name	VW	0.25 [3.52]	0.32 [4.66]	0.32 [4.67]	0.27 [3.92]	0.26 [3.78]	0.23 [3.31]
Quintile	Cap	VW	0.23 [3.57]	0.27 [4.26]	0.28 [4.31]	0.22 [3.44]	0.19 [2.97]	0.16 [2.43]
Decile	NYSE	VW	0.27 [2.81]	0.35 [3.63]	0.36 [3.66]	0.32 [3.25]	0.28 [2.83]	0.26 [2.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.19 [2.69]	0.24 [3.45]	0.23 [3.42]	0.21 [3.11]	0.17 [2.54]	0.15 [2.30]
Quintile	NYSE	EW	-0.02 [-0.37]					
Quintile	Name	VW	0.19 [2.69]	0.25 [3.73]	0.25 [3.70]	0.22 [3.31]	0.20 [2.98]	0.18 [2.68]
Quintile	Cap	VW	0.18 [2.79]	0.23 [3.51]	0.23 [3.53]	0.20 [3.08]	0.16 [2.44]	0.13 [2.09]
Decile	NYSE	VW	0.21 [2.09]	0.28 [2.85]	0.28 [2.86]	0.26 [2.65]	0.21 [2.18]	0.20 [2.01]

Table 3: Conditional sort on size and LGD

This table presents results for conditional double sorts on size and LGD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LGD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LGD and short stocks with low LGD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1973:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LGD Quintiles					LGD Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.36]	0.88 [3.26]	0.96 [3.50]	0.87 [3.13]	0.79 [2.85]	0.13 [1.47]	0.17 [1.90]	0.16 [1.79]	0.11 [1.17]	0.09 [0.96]	0.05 [0.61]
	(2)	0.72 [2.68]	0.90 [3.59]	0.86 [3.40]	0.83 [3.41]	0.89 [3.48]	0.17 [2.22]	0.20 [2.62]	0.17 [2.20]	0.17 [2.21]	0.11 [1.44]	0.12 [1.53]
	(3)	0.78 [3.12]	0.80 [3.65]	0.80 [3.28]	0.83 [3.70]	0.90 [3.84]	0.12 [1.56]	0.17 [2.24]	0.17 [2.17]	0.12 [1.59]	0.15 [1.87]	0.12 [1.48]
	(4)	0.70 [3.03]	0.80 [3.89]	0.87 [3.93]	0.71 [3.40]	0.90 [4.19]	0.20 [2.74]	0.25 [3.41]	0.24 [3.19]	0.21 [2.77]	0.21 [2.71]	0.19 [2.42]
	(5)	0.45 [2.25]	0.62 [3.55]	0.61 [3.08]	0.65 [3.64]	0.77 [4.05]	0.32 [3.73]	0.36 [4.12]	0.38 [4.30]	0.31 [3.56]	0.30 [3.38]	0.26 [2.90]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LGD Quintiles					LGD Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	390	393	393	393	389	37	34	32	33	35	
	(2)	107	107	107	107	107	61	61	60	60	61	
	(3)	76	77	76	76	76	108	109	105	106	109	
	(4)	64	65	64	65	64	231	238	230	233	231	
(5)	59	59	59	59	59	1386	2115	1973	2124	1579		

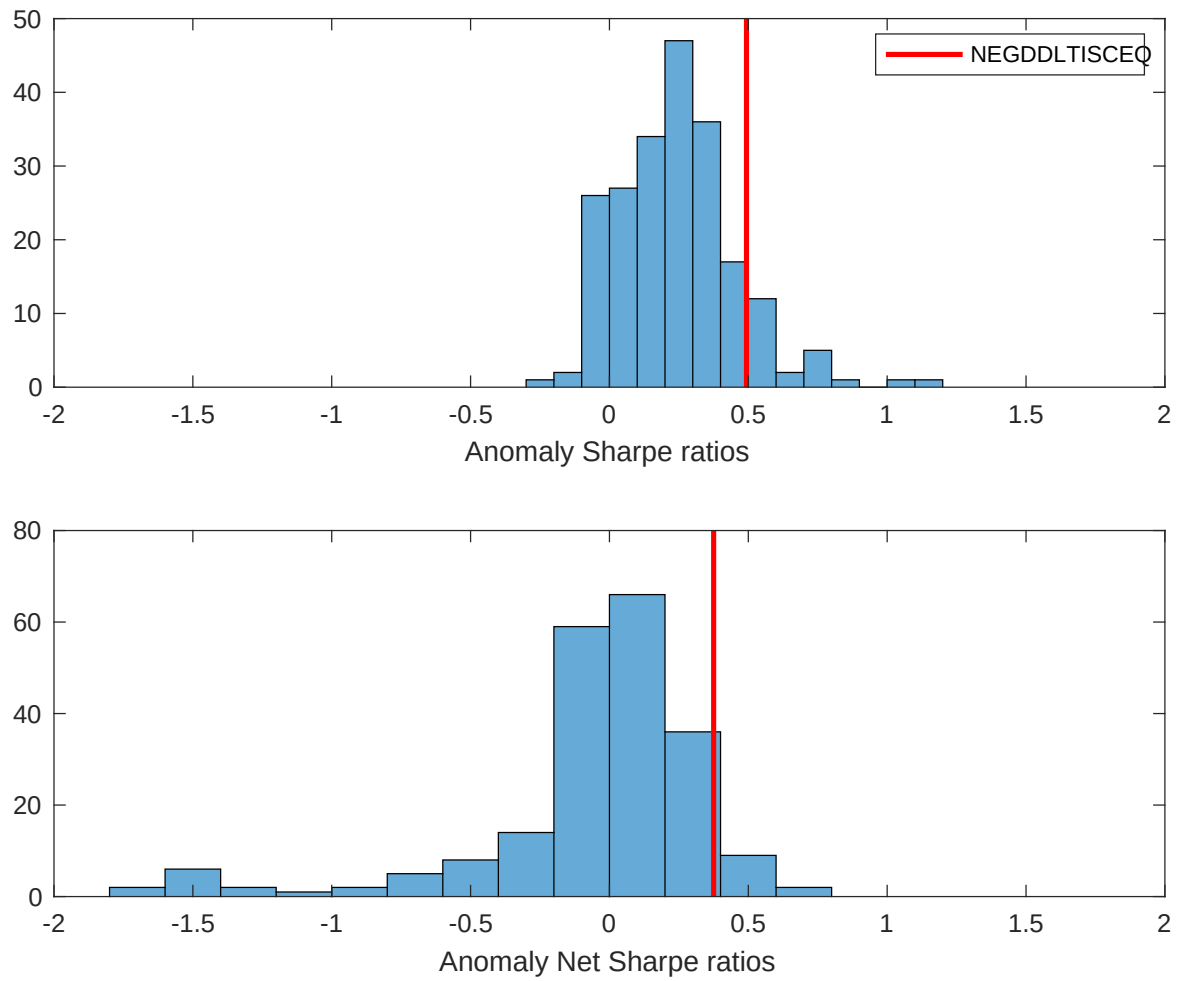


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LGD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

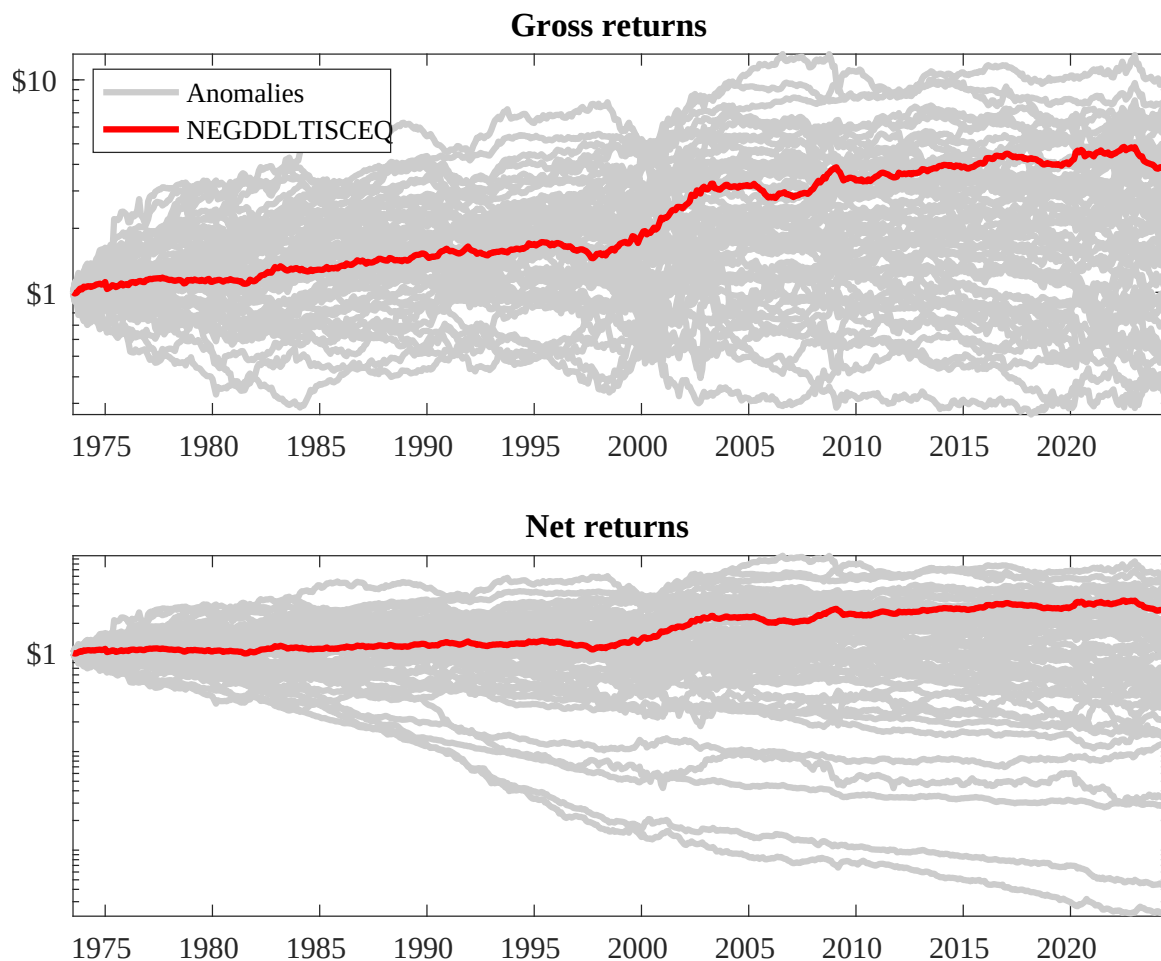
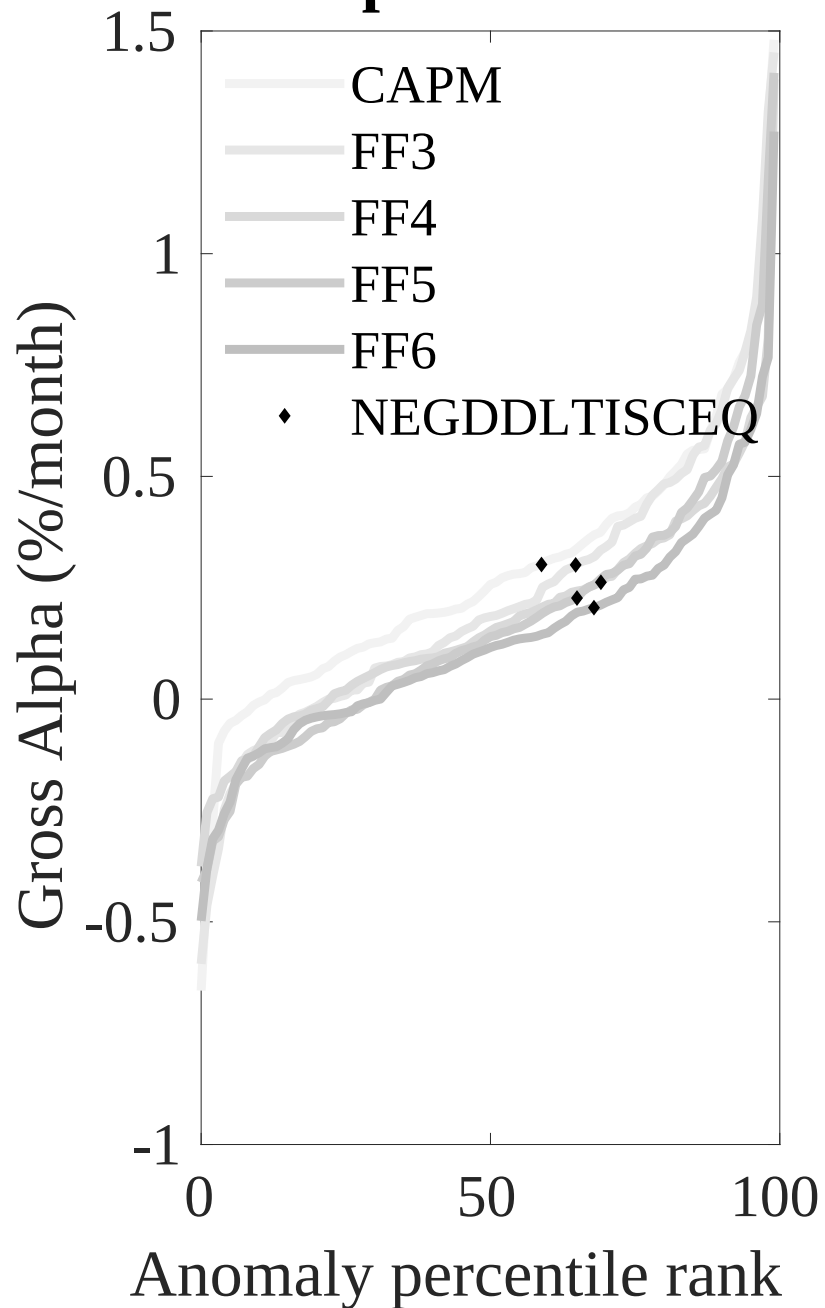


Figure 3: Dollar invested.

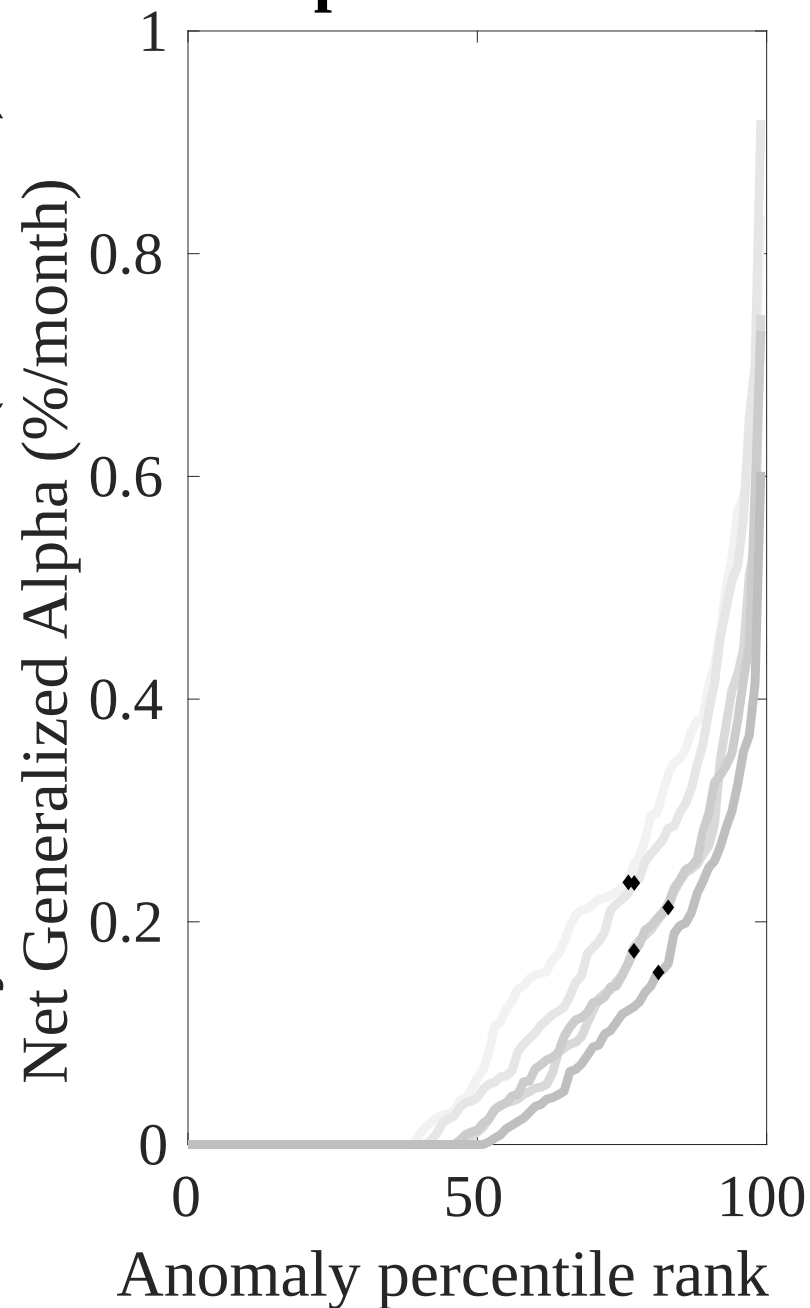
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LGD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



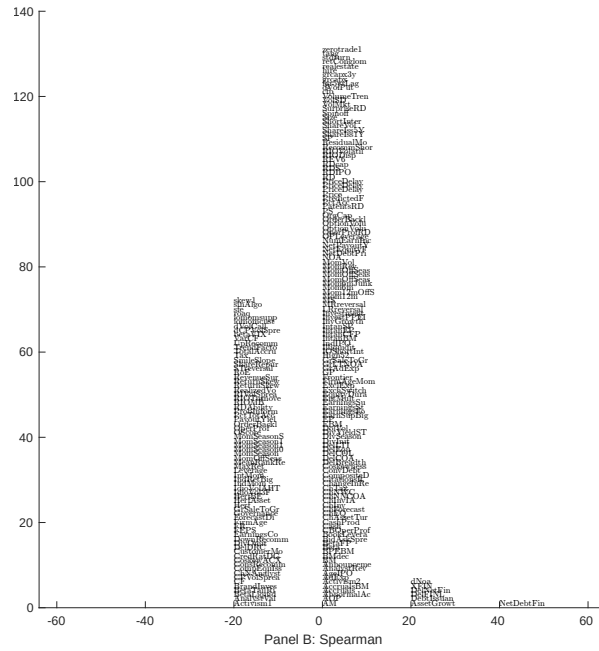
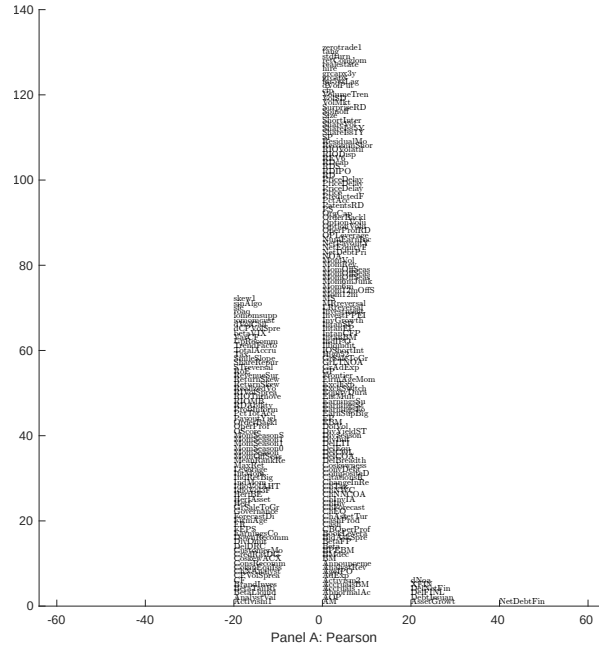


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with LGD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

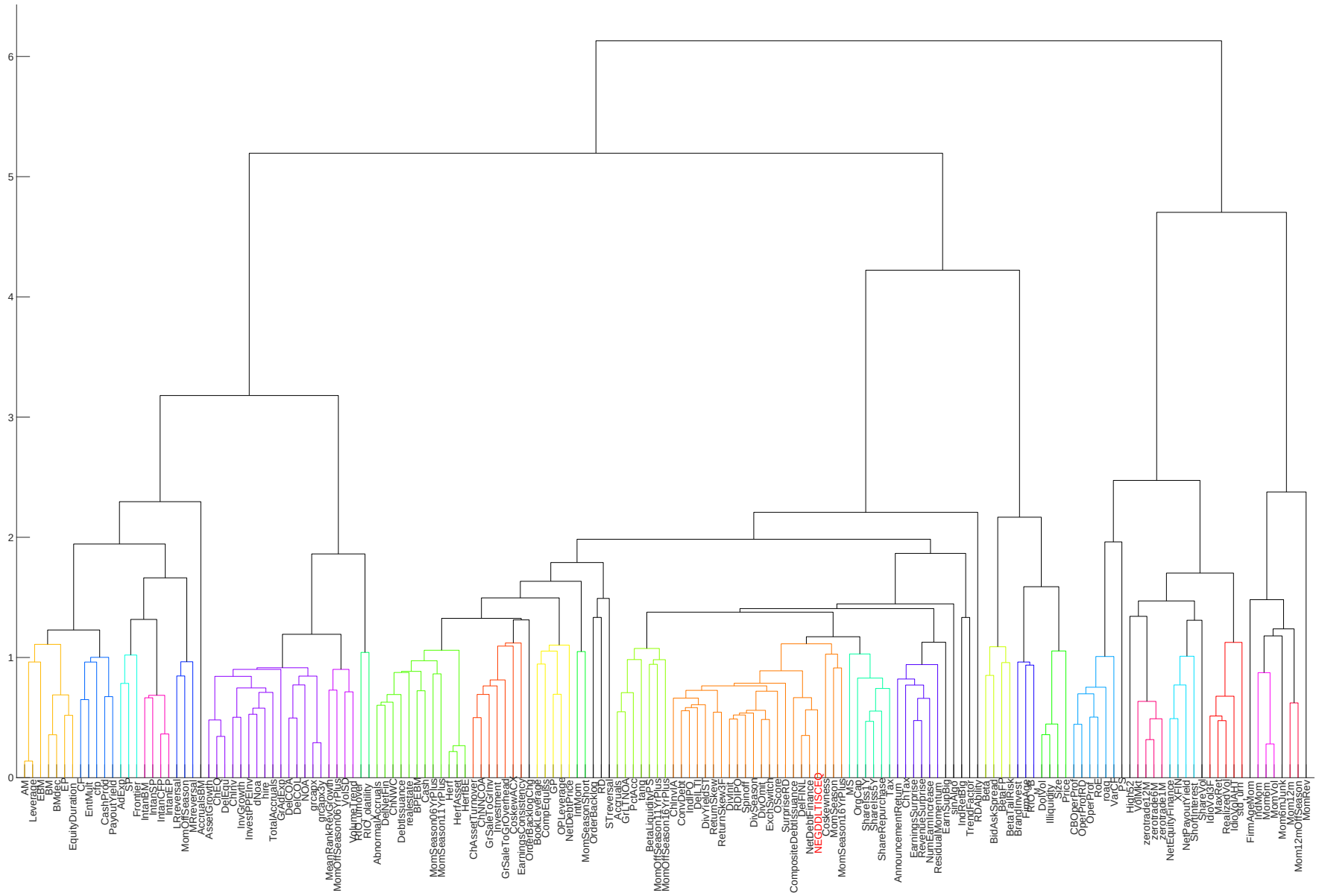


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

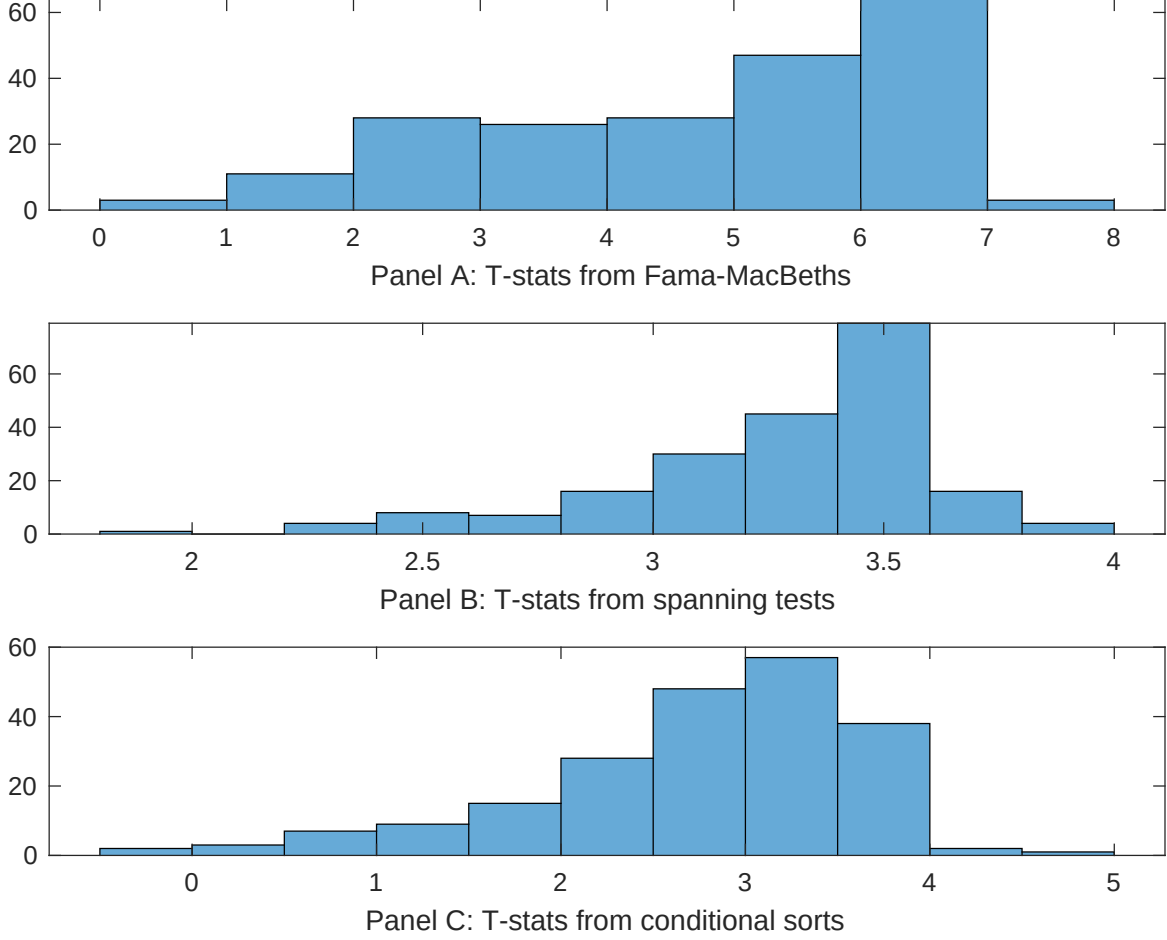


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LGD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LGD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LGD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LGD. Stocks are finally grouped into five LGD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LGD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LGD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LGD}LGD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.68]	0.14 [5.79]	0.14 [5.30]	0.14 [5.65]	0.15 [5.69]
LGD	0.46 [1.89]	0.56 [2.18]	0.80 [2.91]	0.38 [1.50]	0.12 [4.35]	0.33 [1.41]	0.15 [0.50]
Anomaly 1	0.17 [9.30]						-0.11 [-2.23]
Anomaly 2		0.20 [8.75]					0.94 [1.42]
Anomaly 3			0.19 [6.30]				0.10 [1.71]
Anomaly 4				0.10 [8.66]			0.40 [1.78]
Anomaly 5					0.42 [7.13]		0.45 [0.82]
Anomaly 6						0.14 [9.82]	0.94 [5.20]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LGD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LGD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.17 [2.59]	0.18 [2.65]	0.18 [2.62]	0.19 [2.81]	0.20 [2.91]	0.19 [2.71]	0.17 [2.64]
Anomaly 1	17.32 [4.55]						10.33 [1.97]
Anomaly 2		19.58 [5.36]					10.99 [2.20]
Anomaly 3			12.88 [3.80]				8.85 [2.46]
Anomaly 4				6.93 [1.75]			2.48 [0.59]
Anomaly 5					7.30 [2.76]		6.86 [2.55]
Anomaly 6						4.48 [1.15]	-6.09 [-1.40]
mkt	-4.80 [-3.10]	-5.16 [-3.36]	-3.43 [-2.12]	-5.21 [-3.32]	-5.33 [-3.41]	-5.18 [-3.30]	-3.86 [-2.40]
smb	-1.03 [-0.43]	-0.98 [-0.42]	4.89 [1.89]	0.25 [0.10]	1.55 [0.65]	0.83 [0.35]	2.02 [0.75]
hml	-10.89 [-3.64]	-11.24 [-3.79]	-10.60 [-3.51]	-12.35 [-4.08]	-12.26 [-4.07]	-12.41 [-4.07]	-9.76 [-3.24]
rmw	-0.66 [-0.22]	-1.25 [-0.41]	-6.80 [-1.86]	0.93 [0.30]	1.96 [0.63]	0.97 [0.32]	-5.62 [-1.53]
cma	24.04 [5.11]	24.76 [5.38]	21.43 [4.24]	21.43 [3.24]	23.35 [4.54]	26.52 [4.90]	13.20 [1.96]
umd	2.08 [1.32]	2.31 [1.50]	3.49 [2.26]	3.72 [2.39]	2.83 [1.79]	3.43 [2.19]	1.45 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	15	16	15	13	14	13	18

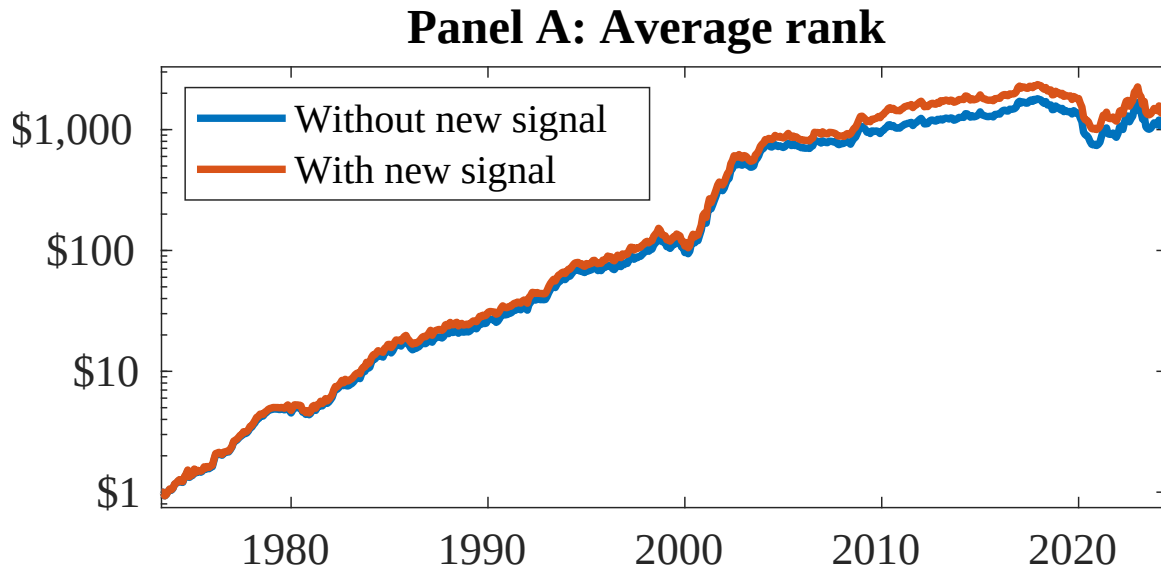


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LGD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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